

Guru Nanak Dev Engineering College, Ludhiana			
Department of Information Technology			
Program	B.Tech.(IT)	Semester	6
Subject Code	PEIT-108	Subject Title	Big Data Analytics
Mid Semester Examination (MSE) No.	1	Course Coordinator	Prof. Himani Sharma
Max. Marks	24	Time Duration	1 hour 30 minutes
Date Of MST		Roll Number	

Note: Attempt all questions

Q.No.	Question	COs,RBT Level	Marks
Q1	Define Metadata and why it is important in Big Data? Give two real -time examples of metadata.	CO1, L1	2
Q2	Distinguish four major differences of quantitative and qualitative analysis in the context of Big Data. Provide examples of techniques used in each type of analysis.	CO1, L4	2
Q3	Describe the different types and elaborate the associated challenges and different Characteristics of Big Data in detail.	CO1, L3	4
Q4	Elucidate the architecture of Hadoop and Apache Spark and elaborate the associated HDFS Commands & Operations of Big Data in detail.	CO4, L2	4
Q5	Compare and Contrast the four major differences of HDFS, GFS and NOSQL storage solutions along with detailed elaboration of different tools for solving big data problems.	CO2, L4	4
Q6	Design and develop the Architectural view of Big Data and also explain different stages associated with Big Data analytics life cycle for business decisions. Elaborate the procedure through appropriate diagram.	CO3, L6	8

Course Outcomes (CO)

1	Understand the concept and challenge of big data
2	Demonstrate hands-on experience on large-scale analytics tools to solve some open BDA problems
3	Examine the impact of big data for business decisions and strategy
4	Apply the novel architectures and platforms like Hadoop introduced for Big data
5	Implement the analytics techniques on a variety of complex applications
6	Apply non-relational databases, for storing and processing large volumes of structured and unstructured data, as well as streaming data

RBT Classification	Lower Order Thinking Levels (LOTS)			Higher Order Thinking Levels (HOTS)		
	L1	L2	L3	L4	L5	L6
RBT Level Number						
RBT Name	Remembering	Understanding	Applying	Analyzing	Evaluating	Creating

Guru Nanak Dev Engineering College, Ludhiana

Department of Information Technology

Program	B.Tech.(IT)	Semester	6
Subject Code	PEIT-108	Subject Title	Big Data A
Mid Semester Examination (MSE) No.	2	Course Coordinator	Prof. Himan
Max. Marks	24	Time Duration	1 hour 30 m
Date Of MST		Roll Number	

Note: Attempt all questions

Q.No.	Question	COs, RBT
Q1	Define the main characteristic of on-disk storage devices and elaborate how on-disk storage devices impact database performance.	CO6, L1
Q2	Distinguish four major differences of Supervised Learning and Unsupervised learning.	CO5, L4
Q3	Describe the architecture and data storage mechanism of MongoDB. Compare collections in MongoDB with tables in RDBMS, and write a basic MongoDB query to insert a document into a collection.	CO4, L3
Q4	Elucidate the various steps involved in implementing CAP Theorem in Big Data.	CO6, L2
Q5	Illustrate the significance of key-Value, Column-Family and Graph database also compare and contrast the four major differences of RDBMS and NoSQL databases .	CO2, L4
Q6	Design and develop the different approaches of Netflix Big Data Analytics for the Emergence of Data Driven Recommendation System in detail along with their challenges. Elaborate the procedure through appropriate diagram.	CO5, L6

Course Outcomes (CO)

1	Understand the concept and challenge of big data
2	Demonstrate hands-on experience on large-scale analytics tools to solve some
3	Examine the impact of big data for business decisions and strategy
4	Apply the novel architectures and platforms like Hadoop introduced for Big c
5	Implement the analytics techniques on a variety of complex applications
6	Apply non-relational databases, for storing and processing large volumes of unstructured data, as well as streaming data

RBT Classification	Lower Order Thinking Levels (LOTS)			Higher Order Thinking	
RBT Level Number	L1	L2	L3	L4	L5

RBT Name	Remembering	Understanding	Applying	Analyzing	Evaluating

Please check that this question paper contains 09 questions and 02 printed pages within first ten

[Total No. of Questions: 09]

[Total No. of Pages: 02]

Uni. Roll No. 2203868...

Program: B.Tech. (Batch 2018 onward)

Semester: 6th

Name of Subject: Big Data Analytics

Subject Code: PEIT-108

Paper ID: 17208

Scientific calculator is NotAllowed

Detail of allowed codes/charts/tables etc. NIL

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Why is Hadoop preferred for big data analytics?
- b) How data validation is important before data analysis?
- c) What is the relation of the term "schema-less" with NoSQL.
- d) Explain the concept of "Consistency" in CAP theorem?
- e) *"The combination of replication and sharding may help in fault tolerance and improve performance in a distributed database system"*. Justify the given statement with appropriate example.
- f) Analyze one case where unsupervised learning is preferred over supervised learning.

Part – B

[Marks: 04 each]

- Q2.** Illustrate the differences of structured and unstructured data with examples.
- Q3.** How does "Volume" impact Big Data analytics?
- Q4.** Differentiate between Key-Value and Document databases.
- Q5.** Compare Spark and Hive. Discuss the cluster architecture for Spark.

- Q6. Predict the problems and challenges of applying data analytics and machine learning models in real-time applications.
- Q7. Justify the need to include both quantitative and quality analysis in Big Data strategy.

Part – C

[Marks: 12 each]

- Q8. How is regression different from classification? Why classification is called "supervised" learning? What is the importance of training and testing data in classification?

OR

Analyze the differences between row-based and column-based on-disk storage in big data.

- Q9. Which phase of the Big data analytics life cycle is most significant while developing various applications? Explain each phase with relevant case study.

OR

How does the architecture of Hadoop Distributed File System (HDFS) helps in handling very large files easily?
